Project - Low Level Design

on

Deploy Education App to Kubernetes

DevOps

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1.Introduction

The **Education App** is an AI-powered educational application designed to generate structured, engaging, and visually rich learning content from a simple topic prompt. It leverages advanced Large Language Models (LLMs) and AI-based image generation to enhance conceptual understanding through both textual explanations and visual representations.

The system is built with modern technologies and follows a complete DevOps lifecycle, including containerization using Docker and automated deployment via CI/CD pipelines on AWS EC2. This ensures scalability, reliability, and real-world production readiness.

The application is accessible through a web interface and supports real-time content generation, making it useful for students, educators, and self-learners.

1.1 Scope of the Document

This document provides a detailed overview of the **design, functionality, and deployment** of the Multimodal Education Creator application.

It covers:

- System architecture and workflow
- Features and functionalities
- Technology stack used
- Deployment process on AWS EC2
- Docker and CI/CD pipeline integration
- Environment configuration

The document is intended to serve as a reference for:

- Developers maintaining or extending the system
- DevOps engineers handling deployment and automation
- Users who want to understand system capabilities

1.2 Intended Audience

This document is intended for the following audience:

- **Developers**
  To understand the system architecture, backend logic, and integration of AI services.
- **DevOps Engineers**
  To manage deployment, CI/CD pipelines, Docker containers, and AWS infrastructure.
- **Educators & Students**
  To understand how the application generates learning content and how it can be used effectively.
- **Project Evaluators / Recruiters**
  To assess the technical depth, real-world implementation, and scalability of the project.

1.3 System Overview

The Multimodal Education Creator is a cloud-deployed AI application that transforms a user-provided topic into structured educational content along with AI-generated visuals.

Core Workflow

1. User inputs a topic via the Streamlit interface
2. The request is sent to the Gemini API for text generation
3. The generated content is processed into structured learning sections
4. A prompt is created for image generation
5. Hugging Face API generates relevant visuals
6. The output (text + images) is displayed in the UI

System Architecture Overview

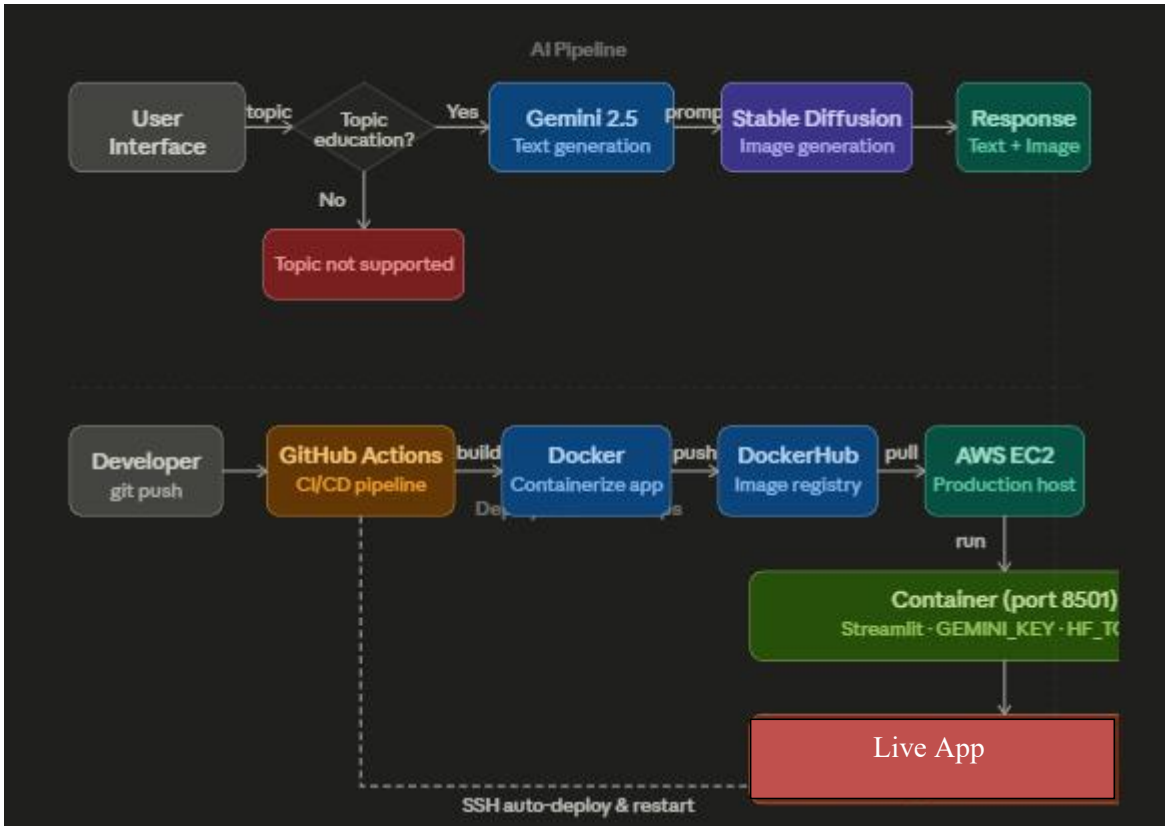
The system follows a **client-server architecture** deployed on the cloud:

- **Frontend (Client Layer)**:
  Built using Streamlit, provides an interactive user interface
- **Backend (Application Layer)**:
  Developed in Python, handles business logic and API communication
- **AI Services (Processing Layer)**:
  - Gemini API for text generation
  - Hugging Face API for image generation
- **Deployment Layer**:
  - Docker for containerization
  - AWS EC2 for cloud hosting
  - GitHub Actions for CI/CD automation

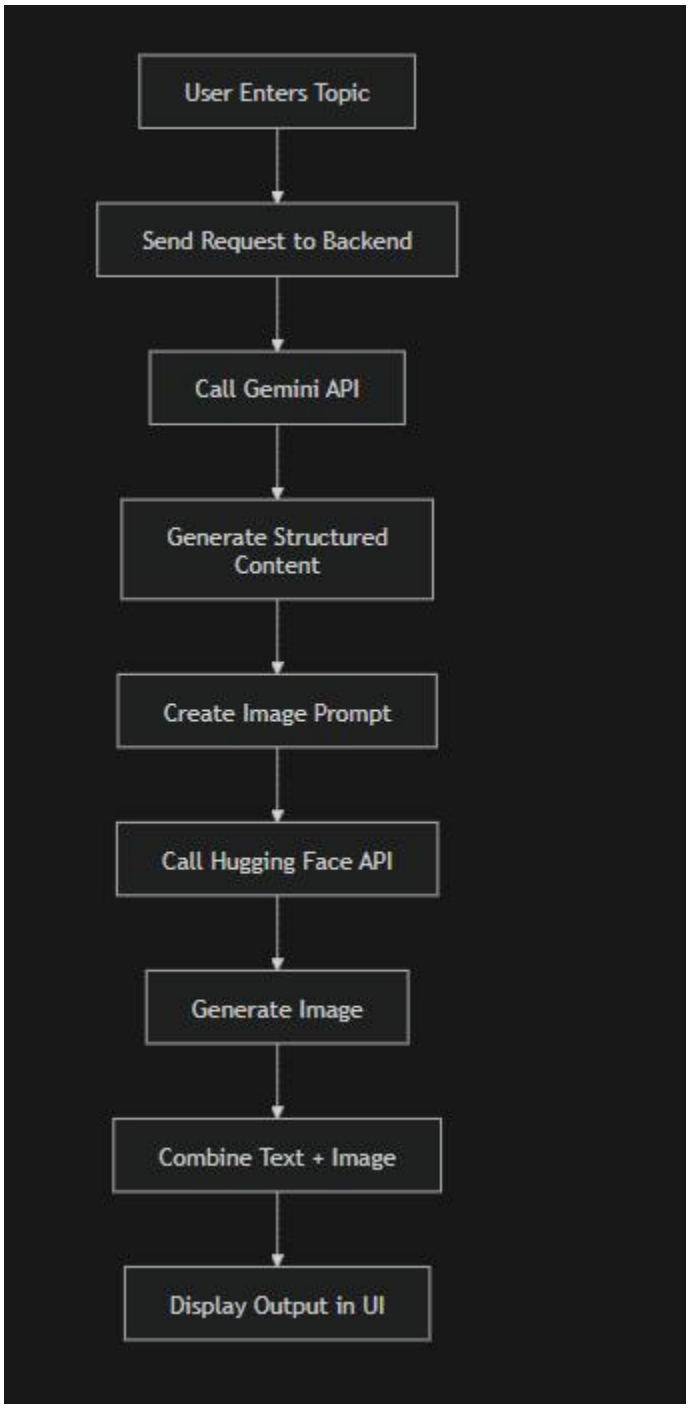
2. System Design

The system is designed as a modular, scalable, and cloud-deployable AI application. It follows a microservice-inspired architecture with clear separation between UI, AI services, and deployment layers.

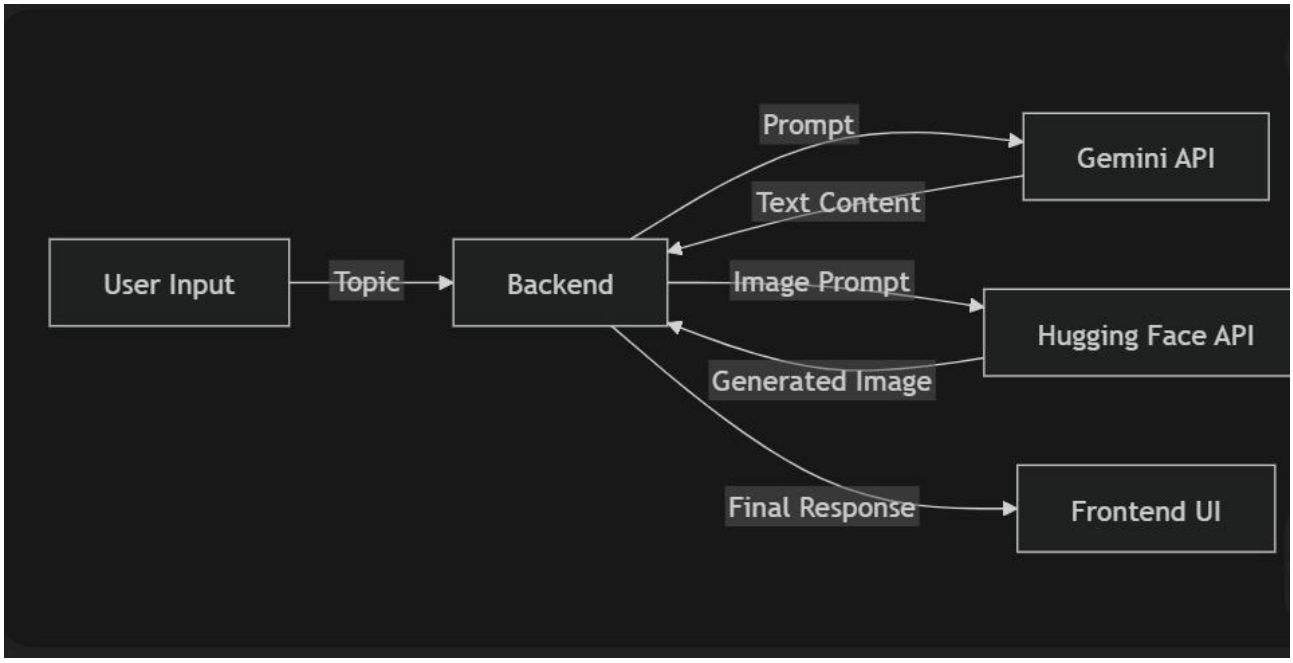
2.1 Application Design



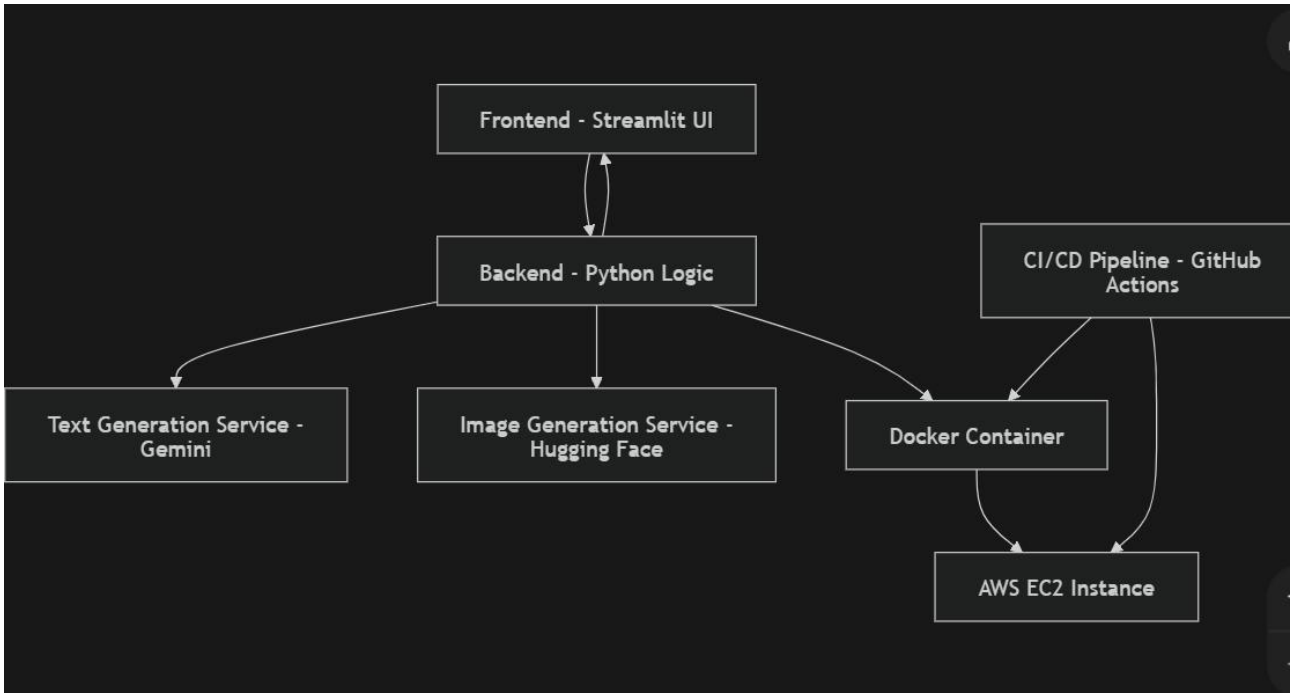
2.2 Process Flow



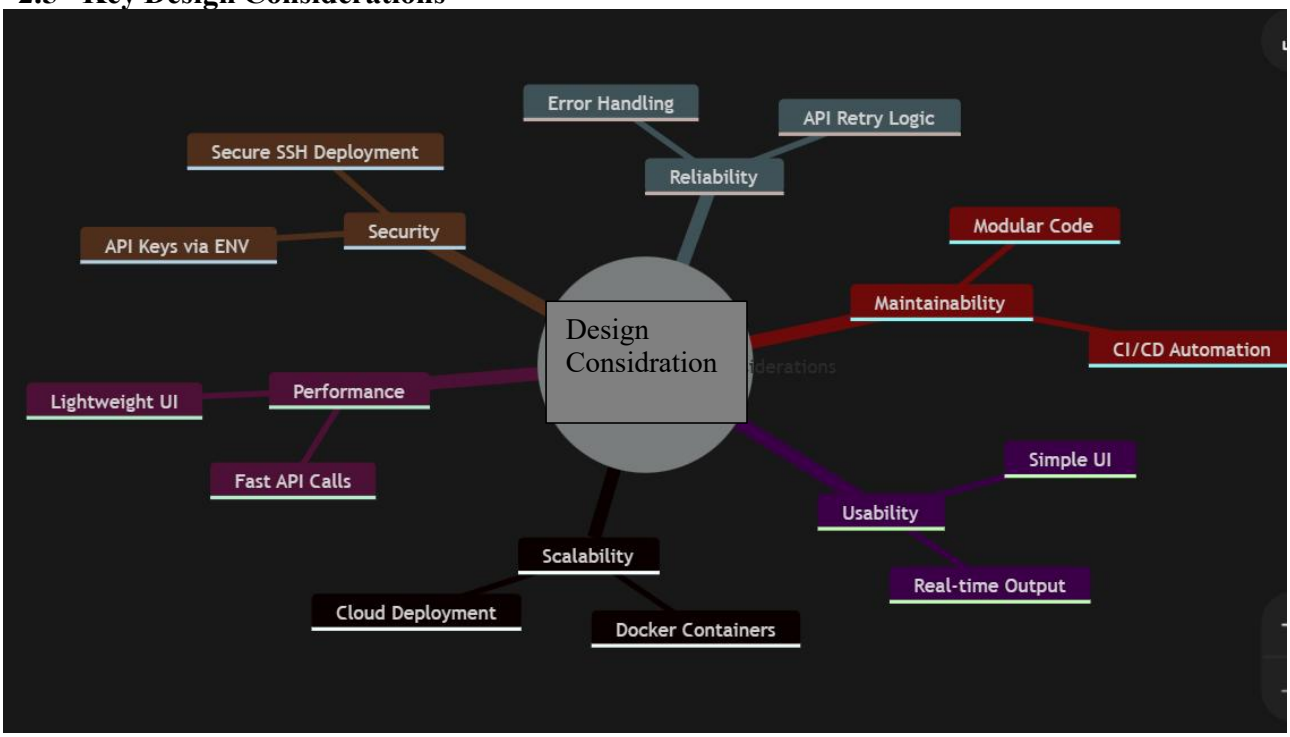
2.3 Information Flow



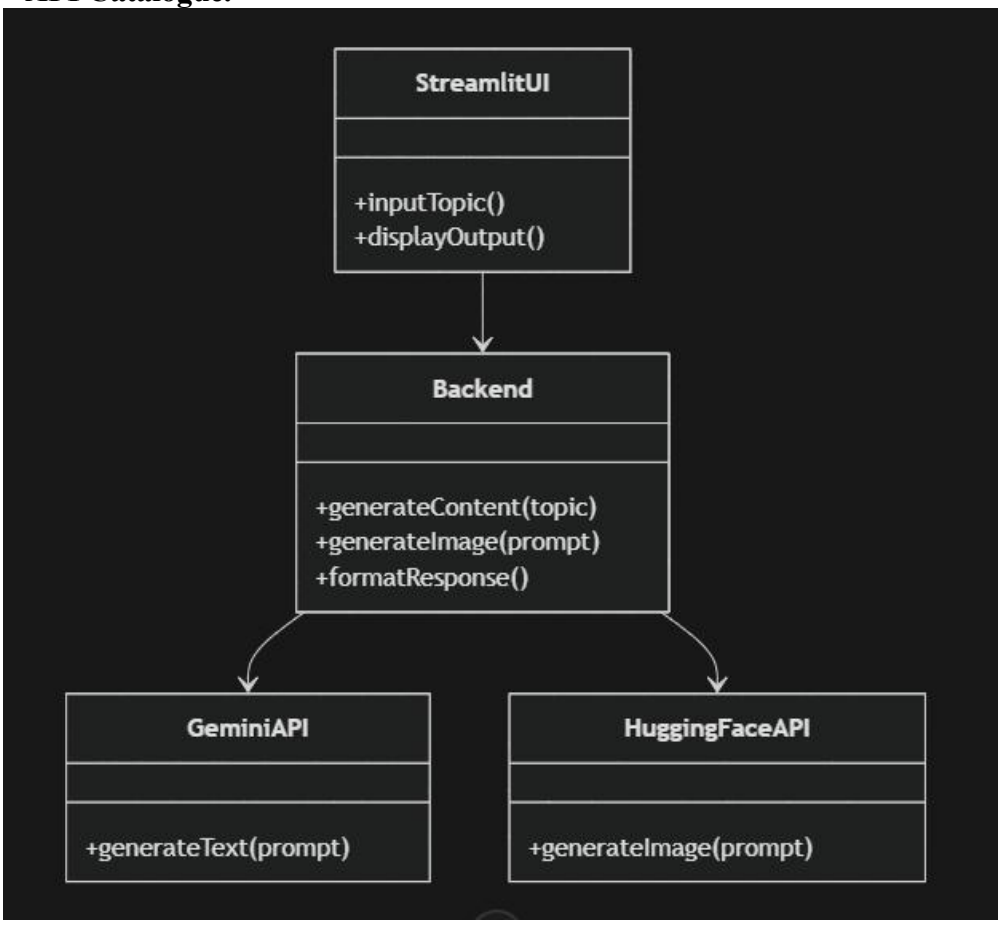
2.4 Components Design



2.5 Key Design Considerations



2.6 API Catalogue.



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DevOps & Deployment

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